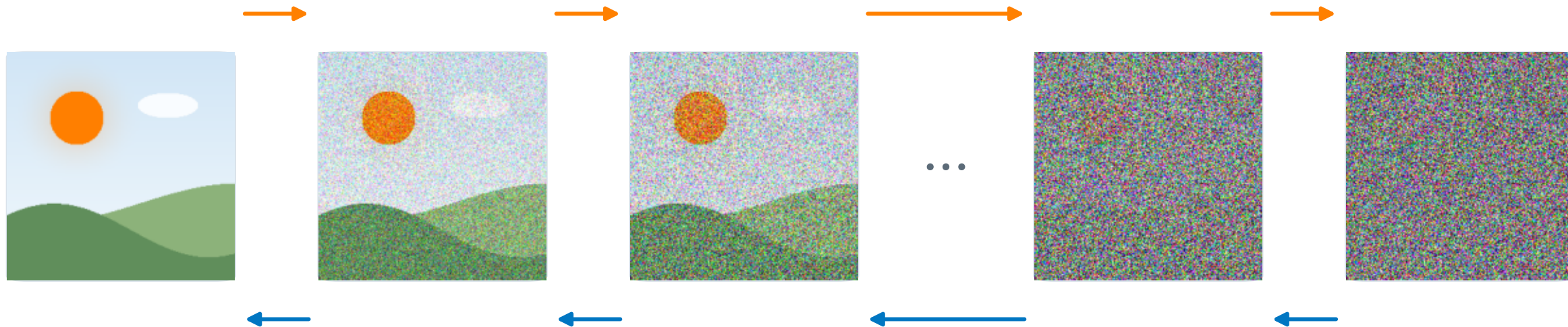


Diffusion models for images: destroy an image with noise, learn to reverse it

Forward process $q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$

fixed schedule — gradually add a little Gaussian noise



x_0

real image

$x_0 \sim q(x_0)$

x_1

x_2

x_{T-1}

x_T

pure noise

$x_T \sim \mathcal{N}(0, \mathbf{I})$

Reverse process $p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$

learned by a neural network — predict the noise and remove it